

R. AKSHAI KUMAR

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SUMMARY:

Innovative and detail-oriented bioinformatics professional with a strong background in computational biology, genomics, data analysis and computational molecular docking. Experienced in leveraging advanced bioinformatics tools to analysis complex biological data, uncover insights, and drive research forward, with basic knowledge in differential expression analysis using DESeq2. Key achievements in phylogenetic analysis, proficiency in R and Python for biological data analysis, visualization and the development of machine learning workflows. Passionate about bridging biology and technology to address real-world challenges and advance scientific discovery.

EDUCATION:

- Bharathidasan university, Trichy
5th year M.sc (Integrated)Bioinformatics with CGPA 8.07 Present
 - Theni kammavar sangam matriculation Higher Secondary School, Theni with 93%
2015 - 2022
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CERTIFICATION:

- Data Analysis with R programming by Google (Coursera).
 - Molecular Biological Techniques, National collage.
 - Data Analysis using R Programming, SANATAN DHARMA COLLEGE AMBALA CANTT.
 - Introduction to Data Science and scikit-learn in Python by LearnQuest (Coursera).
 - Online workshop on AI-Assisted Drug Designing, Barcode Biotechnology.
 - Machine learning with Python by IBM(Coursera)
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SKILLS:

1. Technical Skills:
 - Programming: R (DESeq2), Python.
 - Machine learning
 - Handling Bioinformatics analytic tools and databases.
2. Soft Skills:
 - Communication
 - Team Work
 - Problem Solving
3. Languages:
 - Tamil
 - English

PROJECT:

- **MINI-PROJECT**

Department of Bioinformatics, Bharathidasan University|GitHub: <https://github.com/R-Akshai-kumar02/Academic-Project.git>

- Conducted comparative genomic analysis of 10 major neurotransmission pathways (glutamatergic, GABAergic, dopaminergic, cholinergic, serotonergic, and others) across 17 taxa spanning archaea to vertebrates
- Performed sequence homology assessment and phylogenetic reconstruction to trace evolutionary conservation of neurotransmitter biosynthetic enzymes, transporters, and receptor subunits
- Identified conserved core signaling components in bilaterians (e.g., GAD1, SLC6A4, HTR2A) alongside lineage-specific receptor diversification in vertebrates, particularly primates and findings provide a molecular framework for understanding the evolutionary origins of human neural signaling

- **RESEARCH-INTERN**

Bioinformatics Centre, Madras Veterinary College (MVC), TANUVAS|GitHub:<https://github.com/R-Akshaikumar02/HuntingtonDisease>

- Investigated phytochemicals from *Juglans regia* L. (walnut) as potential inhibitors of mutant huntingtin (mHTT) protein aggregation using computational drug discovery approaches
- Screened 209 phytochemicals from the BIMP database for pharmacokinetic and toxicity properties using SwissADME and ProTox-3.0, and predicted pharmacodynamic activity via PASS online server
- Performed comparative molecular docking of 6 shortlisted compounds against huntingtin protein using AutoDock Vina and LibDock (BIOVIA Discovery Studio)
- Identified Geranic acid as the lead compound with the most favorable binding near the polyQ region; validated complex stability through molecular dynamics simulation using OpenMM

- **PERSONAL PROJECT**

BioHTT-tool ML-Based Bioactivity Predictor for Huntingtin (HTT) Protein

- Developed and deployed a Streamlit web application to predict compound bioactivity (Active/Inactive) against the Huntingtin (HTT) protein, supporting Huntington's disease drug discovery.
- Processed SMILES into Morgan fingerprints (ECFP4, 2048-bit) using RDKit and trained a Random Forest classifier to predict bioactivity with confidence scores.
- Implemented QSAR descriptor calculation and integrated the trained model into a user-friendly interface for real-time compound screening and prediction.

GitHub:<https://github.com/R-Akshai-kumar02/BioHTT-Tool>

Tool Link: <https://biohtt-tool-a9plsplwkqbevgoore5bir.streamlit.app/>